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EXPERIMENT-4

To build an Artificial Neural Network using the Backpropagation algorithm and test it using a dataset.

Python Program:

```
import pandas as pd

from google.colab import files

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score, classification_report


# Upload CSV file

uploaded = files.upload()


# Read dataset

filename = list(uploaded.keys())[0]

data = pd.read_csv(filename)


print("Dataset:")

print(data.head())


# Convert categorical columns into numeric
```

```
encoders = {}
for col in data.columns:
    if data[col].dtype == 'object':
        le = LabelEncoder()
        data[col] = le.fit_transform(data[col])
        encoders[col] = le

# Features and Target
X = data.iloc[:, :-1]
y = data.iloc[:, -1]

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create ANN using Backpropagation
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation='relu',
    solver='adam',
```

```
    max_iter=1000,
    random_state=42
)

# Train the model
model.fit(X_train, y_train)

# Test the model
y_pred = model.predict(X_test)

# Accuracy
print("\nAccuracy:", accuracy_score(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Predict a new sample (Example)
new_sample = X.iloc[[0]]
new_sample = scaler.transform(new_sample)

prediction = model.predict(new_sample)

# Convert prediction back to original label if target was encoded
target_col = data.columns[-1]
if target_col in encoders:
```

```
    prediction = encoders[target_col].inverse_transform(prediction)


print("\nPrediction for New Sample:")

print(prediction[0])
```

Output:

```
ANN_Buys_Computer_Dataset.csv(text/csv) - 539 bytes, last modified: 7/29/2026 - 100% done
Saving ANN_Buys_Computer_Dataset.csv to ANN_Buys_Computer_Dataset.csv
Dataset:
     Age  Income Student Credit_Rating Buys_Computer
0   <=30    High      No          Fair            No
1   <=30    High      No     Excellent            No
2  31-40    High      No          Fair           Yes
3    >40  Medium      No          Fair           Yes
4    >40     Low     Yes          Fair           Yes

Accuracy: 0.3333333333333333

Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00         3
           1       0.40      0.67      0.50         3

    accuracy                           0.33         6
   macro avg       0.20      0.33      0.25         6
weighted avg       0.20      0.33      0.25         6
```